

NLP for Teens

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■ GenAI

GenAI (Generative AI)

Goal: create new content

Tasks:

- text generation

- image generation

- code generation

- audio/video generation

GenAI = creating new content.

■ GenAI

↔ How they connect (super short)

- NLP → made embeddings
- NLP → made transformers
- Transformers → became LLMs
- LLMs → became GenAI

Chain:

NLP → Transformers → LLMs → GenAI

Meaning:

Each step builds the next one.

■ Transformers

Transformers are neural networks that understand relationships between words using attention, not recurrence.

They read all words at the same time, not one-by-one like RNN/LSTM.

Transformers introduced self-attention, which lets every word look at every other word.

■ Transformer model types

1. Encoder-only → understanding

BERT, RoBERTa, DistilBERT (BERT = pre-trained on huge text (Wikipedia + Books).)

BERT = original

RoBERTa = stronger

DistilBERT = smaller & faster

Used for: classification, NER, sentiment, embeddings

2. Decoder-only → text generation

GPT, LLaMA

Used for: chatbots, writing, code

3. Encoder-Decoder → translation, summarization

T5, BART (Used for: seq-to-seq tasks)

T5 = unified text-to-text system

BART = BERT-encoder + GPT-decoder

■ BART

BART = Bidirectional + Autoregressive + Transformer

Bidirectional → like BERT

Reads the whole input at once (encoder)

Autoregressive → like GPT

Generates output one token at a time (decoder)

So:

BART = BERT-style encoder + GPT-style decoder

Assignment

- Use a transformer model to summarize the following text.
 - See next slide for the text.
- Tasks
 - Load_a_summarization_model** using the `transformers` library.
 - Generate_a_summary** (2–3 sentences).

Assignment Text

- Climate change is one of the most significant challenges facing humanity today. Rising global temperatures have led to melting ice caps, extreme weather events, and disruptions in ecosystems around the world. Governments, scientists, and communities are working together to find sustainable solutions, reduce carbon emissions, and protect the planet for future generations.

Summary

- ✓ Transformer_models: encoder–decoder architecture (BERT vs GPT).
- ✓ LLM_tasks: using transformers for summarization, translation, and text generation.
- ✓ Summarization_pipeline: loading a transformer model to produce short summaries.
- ✓ Assignment_overview: summarize a given text using a transformer model.

Thank You

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